1. What is the main purpose of using a loop structure in a program?
To repeat a block of code over and over so you don't have to copy and paste the same lines a bunch of times. It just keeps running until a certain condition stops it.

2. Explain the difference between a while loop and a do-while loop.
A while loop checks the condition at the very start before it even runs the code once. A do-while loop runs the code inside first, and then checks the condition at the bottom. This means a do-while loop always runs at least one time no matter what.

3. What application or section does the textbook reference regarding input verification?
The Prompter application section.

4. Loop Errors and Terms:
a) What is an infinite loop? It's a loop that gets stuck and never stops running because the exit condition never becomes false.
b) Give two examples of errors that cause an infinite loop. One is putting a accidental semicolon right after your while condition line. Another is forgetting to change your loop variable inside the brackets so it stays the same forever.
c) What does the term "overflow" mean? It's when a number gets way too big for the computer's memory space to hold, so it breaks or glitches out.

5. If a loop is set up to double a variable starting at 1 while it's less than 50, what value makes the loop terminate?
The value is 64. The loop goes 1, 2, 4, 8, 16, 32, and then it doubles to 64. Since 64 is not less than 50, it stops right there.

6. If a loop decreases a variable by 5 while it's not equal to 0, will it always stop or could it become an infinite loop?
It could easily become an infinite loop. If the number starts at something that isn't a multiple of 5 (like 3 or 7), or if it starts as a negative number, subtracting 5 will just skip 0 entirely and keep going forever.

7. Compare a counter variable and an accumulator variable. How are they different?
A counter just counts how many times something happens and always goes up by 1, like count = count + 1. An accumulator adds up a running total of changing numbers, like total = total + score.

8. Convert the following math statement into a standard student-style for loop:

```
for (int i = 3; i <= 10; i++) {
total = total + i;
}
```

9. What are two things you should look at when deciding which loop structure (for, while, or do-while) is best to use?

First, see if you already know exactly how many times the loop needs to repeat (if you know, use a for loop). Second, see if the code has to run at least once before checking the rule (if it does, use a do-while loop).

11. String Methods Analysis Outputs:

Suppose the code has a line like: String word = "my string.";

a) What does word.length() return?

Answer: 10

b) What does word.substring(0, 2) return?

Answer: my

c) What does word.substring(0) return?

Answer: my string.

d) What does word.toUpperCase() return?

Answer: MY STRING.

e) What does word.toLowerCase() return?

Answer: my string.